RECENT TRENDS PRESENTATION

AirLLM

Running 70B-parameter LLMs on a 4 GB GPU, one layer at a time

Loading 80 decoder layers · 62%

ATMIYA UNIVERSITY
RAJKOT, GUJARAT - INDIA

A GRADE
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PRESENTED BY

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Recent Trends Presentation

70B	4 GB	80
parameters	GPU memory	decoder layers

Agenda

Eight modules in this talk

03 The Memory Wall

04 Key Insight: Spatial → Temporal

05 Architecture: Split · Store · Stream

06 The Inference Loop

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KEY QUESTION

How can a 70-billion-parameter model run on a GPU with only 4 GB of memory?

Short answer: stream the model layer by layer instead of loading it all at once.

The Memory Wall

Model weights have outgrown consumer GPUs



70B model
≈ 140 GB in fp16

ERROR

CUDA out of memory

Needed: 140 GB
Available: 4 GB

The model never finishes loading.

Spatial memory

Traditional loading keeps every layer resident at the same time, so GPU memory must hold the entire model.

Weight size in fp16

parameters × 2 bytes

Model	fp16 weights
7B	14 GB
13B	26 GB
70B	**140 GB**
405B	810 GB

Consumer GPUs

4–24 GB VRAM

A 70B model needs 35× a 4 GB card

Key Insight: Spatial → Temporal

Decoder layers run one after another, so memory can be reused over time

SPATIAL · TRADITIONAL

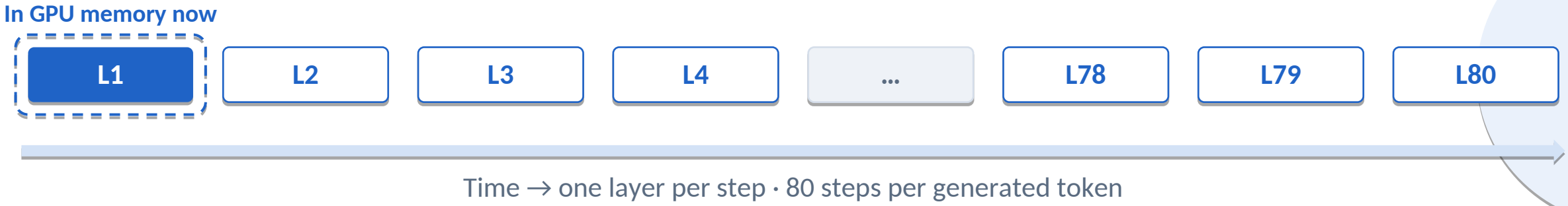
Load all 80 layers, then compute.

GPU memory ≈ all layers ≈ 140 GB

TEMPORAL · AIRLLM

Stream one layer in, compute, evict, repeat.

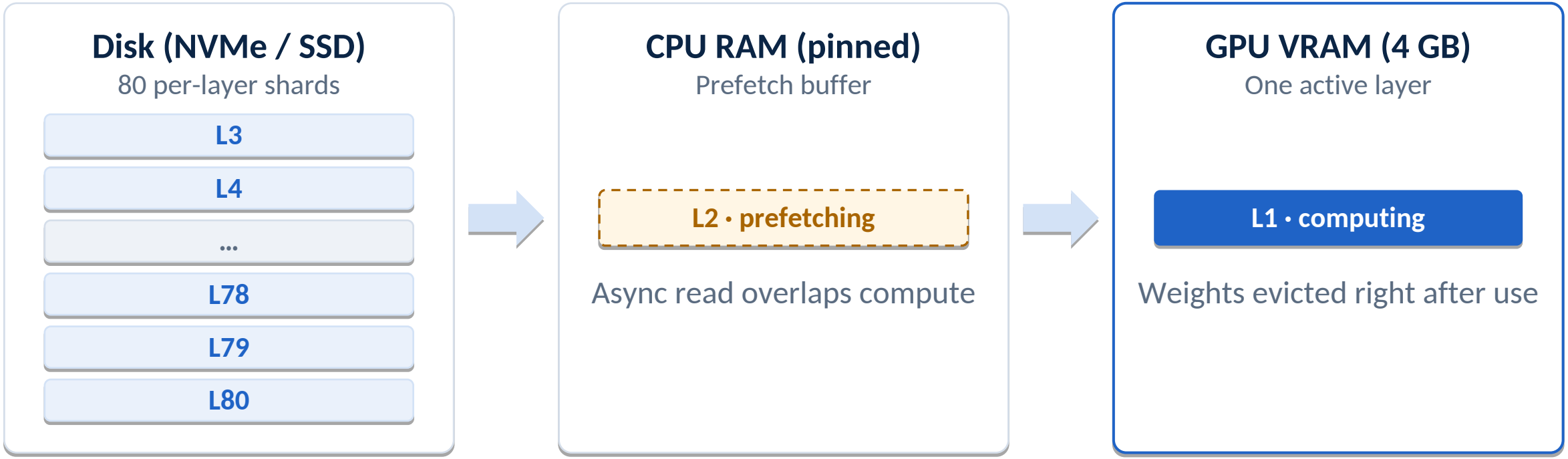
GPU memory ≈ one layer ≈ 1.7 GB*



* fp16 weights of one 70B decoder layer (≈0.86B parameters × 2 bytes); activations, KV cache and CUDA context add overhead.

Architecture: Split · Store · Stream

Split once, then stream layer shards through three memory tiers

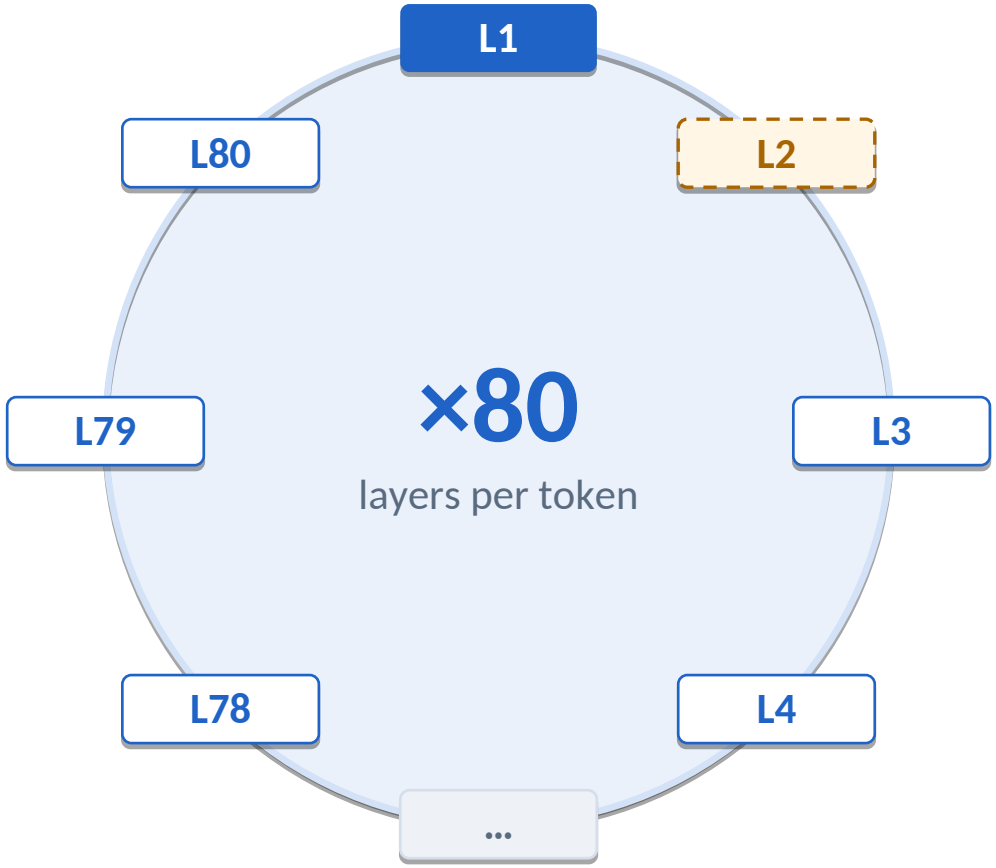


USAGE (PYTHON)

```
>>> from airllm import AutoModel
>>> model = AutoModel.from_pretrained("meta-llama/Llama-2-70b-hf")
>>> out = model.generate(ids.cuda(), max_new_tokens=20)  # streams per layer
```

The Inference Loop

Per generated token, repeated across all 80 layers



01 Compute

Run layer N's attention and MLP on the GPU.

02 Evict

Free layer N's weights so GPU memory stays flat.

03 Prefetch

Read layer N+1 from disk while N computes (≈10% faster, v2.5).

Scaling Up: MoE & Compression

Fewer bytes per token → bigger models (figures: AirLLM README)

2.8T
Kimi K3 · ≈3.72 GB

671B
DeepSeek-V3 · ≈12 GB

70B
Llama-class · ≈4 GB

MoE expert streaming

Kimi K3 routes each token to 16 of its 896 experts; AirLLM streams only those.



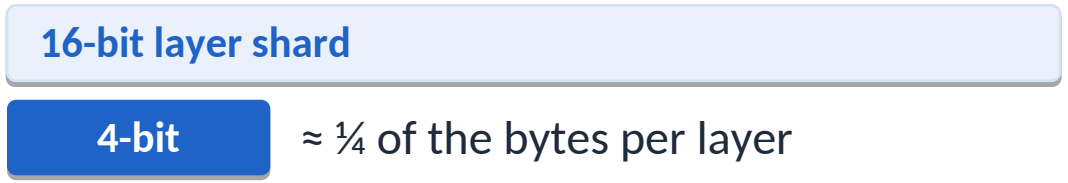
Routed: streamed from disk Idle: stays on disk

Block-wise compression (optional)

`compression='4bit' | '8bit'`

Quantizes the weights only; activations stay 16-bit.

README claims up to 3× faster; issue #330 reports slowdowns.



≈ ¼ of the bytes per layer

Accessibility vs. Latency

Seconds per generated token for a 70B model (indicative)

AirLLM · 16-bit
No VRAM/RAM barrier



~121 s/token

AirLLM · 4-bit
Warm cache · same 4 GB GPU



~25 s/token

llama.cpp · quantized
Requires 44 GB RAM



~3.4 s/token

Use it for

Offline or batch generation, research on very large models, prototyping on laptops.

Avoid for

Interactive chat and latency-critical serving: every token streams all 80 layers again.

Related Trends

Other routes to large models on small hardware

AirLLM
Layer-wise streaming

Streams one layer at a time from disk: minimal VRAM, high latency.

FlexGen
ICML 2023

Offloads weights and KV cache across GPU, CPU and disk for high-throughput batch inference.

ZeRO-Inference
DeepSpeed

Keeps weights in CPU RAM or NVMe and streams them to the GPU on demand.

llama.cpp
GGUF · CPU + GPU

Quantized inference with memory-mapped weights on commodity hardware.

PowerInfer
SOSP 2024

Hot neurons stay on the GPU, cold ones on the CPU, exploiting activation sparsity.

HF Accelerate
device_map="auto"

Spills layers that don't fit onto CPU RAM or disk automatically.

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Thank you

Questions and discussion

KEY TAKEAWAYS

1. GPU memory need drops to roughly one layer, not the whole model.
2. Disk bandwidth becomes the new bottleneck.
3. It trades latency for accessibility: a research and batch tool, not a chat server.

All 80 layers streamed · complete

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